

CS 3.307

Performance Modeling for Computer Systems

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Logistics

- ▶ Feel free to contact me anytime at tejas.bodas@iiit.ac.in.
- ▶ Office @ A5301.
- ▶ Book– Performance modeling and design of computer systems (Cambridge press) by Mor Harchol-Balter (Professor, CMU)
- ▶ Other books: 1) Stochastic processes by Sheldon Ross 2) Probabilistic modeling by Isi Mitrani.
- ▶ Assignment 1 : 15%. Midsem exam: 25%. Project: 25%
Endsem 35 %.

Course Outline

- ▶ Module 1 (2 lectures)
 - ▶ Motivation, Probability refresher, Introduction to Stochastic Processes
- ▶ Module 2 (4 lectures)
Poisson Process & Markov Chains
- ▶ Module 3 (2 lectures) Elementary Queues
- ▶ Module 4 Renewal theorems and Busy period analysis (3 lectures)
- ▶ Module 5 (3 lectures) Advanced Queues

Performance modeling for Computer systems

- ▶ How do you measure the performance of your computer?
- ▶ Speed with which it runs programs. RAM, clock speed, GPU, Cores.
- ▶ Storage space ? SSD or not ?
- ▶ What is the key word here ? LATENCY!
- ▶ Performance metrics?
 - ▶ response time (run time, lag, delay, jitter)
 - ▶ blocking probability (screen freeze, no disk space, packet loss, buffer full)

Modeling ?

- ▶ Design for performance: How many cores or GPU's? which core to use? how to schedule instructions in a core?
- ▶ Routing (which core) and scheduling (which program/instruction to execute)
- ▶ How do you know which is a good design? via experimentation?(costly!)
- ▶ Performance analysis! via stochastic modeling

Applications Beyond Computers

- ▶ Computer systems
 - ▶ server farms, cloud computing, distributed storage systems
 - ▶ Communication systems, Wifi, Sensor networks.
- ▶ Healthcare
 - ▶ How many OT? How many Specialists or nurses?
 - ▶ Scheduling operations, stocking of medicines, scheduling tests.
- ▶ Hospitality industry
 - ▶ Designing hotel lobbies for faster checkin
 - ▶ Restaurant seating! (How many tables of size 2,4,8?)
- ▶ Transportation systems
 - ▶ Airline or Railway scheduling
 - ▶ Priority scheduling, class differentiation
- ▶ Operation Research!
- ▶ Henceforth use the term Queueing system!

A single server queue



- ▶ One server, one FIFO queue for jobs to wait.
- ▶ μ denotes service rate, λ denotes the arrival rate.
- ▶ Service requirements S_n and inter-arrival times A_n are typically assumed to be i.i.d.
- ▶ In its simplest form, we will assume $S_n \sim \text{Exp}(\mu)$ and $A_n \sim \text{Exp}(\lambda)$.
- ▶ Jobs face queueing delay due to waiting for other jobs.
- ▶ This is the most basic $M/M/1$ queue. Modeling this as a Markov chain and solving its stationary distribution gives us mean response time (mean of service time + waiting time).
- ▶ $E[T] = \frac{1}{\mu - \lambda}$.

A single server queue



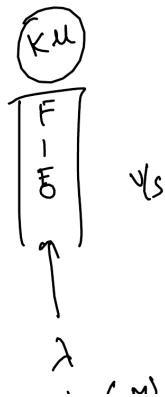
- ▶ $E[T] = \frac{1}{\mu - \lambda}$.
- ▶ Let N is the number of jobs in the system (Queue + server). Then what is $E[N]$?
- ▶ We will see Little's law that says that $E[N] = \lambda E[T]$.
- ▶ Mean number of jobs $E[N] = \frac{\lambda}{\mu - \lambda}$.
- ▶ This course is about Markov chain analysis to derive such formulas.

Example 1: Doubling the arrival rate



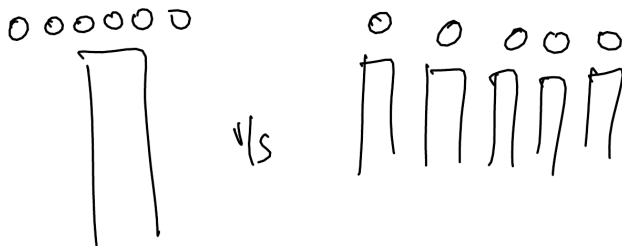
- ▶ $E[T] = \frac{1}{\mu - \lambda}$.
- ▶ What would happen to $E[T]$ if $\lambda \rightarrow 2\lambda$?
- ▶ It could blow up if $\mu < 2\lambda$.
- ▶ If you want to maintain the same level of response time then do you need to double μ ?
- ▶ This course is about making such design choices!

Example 2: A fast server versus many slow servers



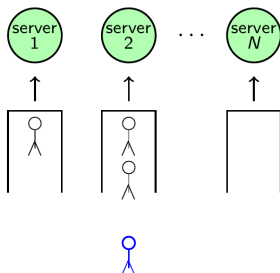
- ▶ Which system will have lower $E[T]$?
- ▶ Is a fast server ($K\mu$) better than K normal servers (μ)?
- ▶ Does job variability impact this decision? Suppose job sizes were XS, S, M, L, XL .
- ▶ In the first model, an S , or M job has to possibly wait behind XL . This is avoided in the second scenario.

Example 3: Central queue or individual



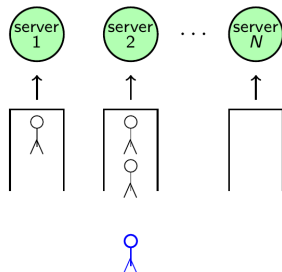
- ▶ At Airport immigration, Hotel check-ins you often see central queues.
- ▶ But at movie theatres, metro/train ticket counters, you see the second model.
- ▶ Which setting has a lower $E[T]$?
- ▶ This course will help you answer such performance modeling questions.

Example 4: Supermarket queue and load balancing



- ▶ Load balancing concerns the questions which queue to join/assign?
- ▶ Popular policy is Join shortest Queue (JSQ).
- ▶ What should be ideally done is Join smallest work (JSW).
- ▶ N is typically large and the overhead in obtaining queue length information is huge ($2N$).

Example 4: Supermarket queue and load balancing



- ▶ In that case, sample d servers randomly and join appropriate queue using $JSQ(d)$ or $JSW(d)$.
- ▶ Problem with JSW or $JSW(d)$ is that the workload information is typically unknown. How to implement it then?
- ▶ How about replicating jobs on d servers and cancelling copies when one copy starts service ?
- ▶ This is redundancy- d with cancel on start.
- ▶ We do this at super-markets all the time!

Probability Refresher

Random experiments and Sample space

- ▶ Random experiment : Experiment involving randomness
 - ▶ Coin toss
 - ▶ Roll a dice
 - ▶ Pick a number at random from $[0, 1]$.
- ▶ Sample space Ω : set of all possible outcomes of the random experiment. It could be a finite or infinite set.
 - ▶ $\Omega_c = \{H, T\}$
 - ▶ $\Omega_d = \{1, 2, \dots, 6\}$
 - ▶ $\Omega_u = [0, 1]$

Events

- ▶ A subset $A \subseteq \Omega$ is called an **event**.
- ▶ Examples of events
 - ▶ Events in the coin experiment: $C_1 = \{T\}$.
 - ▶ Events in the dice experiment: $D_1 = 6, D_2 = \{1, 3, 5\}$
 - ▶ Events in $U[0, 1]$ experiment: $U_1 = \{0.5\}, U_2 = [.25, .75]$.
- ▶ Probability of event A is denoted by $\mathbb{P}(A)$.

Probability theory

{Random experiment, Sample space, Events} are the key ingredients in probability theory.

In probability theory, we are interested in **measuring** the probability of subsets of Ω (events).

Probability measure $\mathbb{P}$ is a **set function**, i.e. it acts on sets and measures the probability of such sets.

sigma-algebra as domain for $\mathbb{P}$

- ▶ Event space or *sigma-algebra* $\mathcal{F}$ is a collection of measurable sets that satisfy
 - $\emptyset \in \mathcal{F}$ • $A \in \mathcal{F} \implies A^c \in \mathcal{F}$
• $A_1, A_2, \dots, A_n, \dots \in \Omega \implies \bigcup_{n=1}^{\infty} A_n \in \Omega$
- ▶ The σ -algebra is said to be closed under formation of compliments and countable unions.
- ▶ Is it also closed under the formation of countable intersections?

When Ω is countable and finite, we will consider power-set $\mathcal{P}(\Omega)$ as the domain.

Formal definition of Probability measure $\mathbb{P}$

Definition

A probability measure $\mathbb{P}$ on the *measurable space* $(\Omega, \mathcal{F})$ is a function $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ s.t.

1. $\mathbb{P}(\emptyset) = 0, \quad \mathbb{P}(\Omega) = 1$
2. For a disjoint collection of event sets $A_1, A_2, \dots$ from $\mathcal{F}$ we have

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$$

(countable additivity)

- The trio $(\Omega, \mathcal{F}, \mathbb{P})$ is called as a probability space.

Conditional probability

- ▶ Given/If dice rolls odd, what is the probability that the outcome is 1?
- ▶ Given/If $\bar{\omega} \in [0, 0.5]$ what is the probability that $\bar{\omega} \in [0, 0.25]$?
- ▶ The conditional probability of event B given event A is defined as $\mathbb{P}(B/A) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)}$ when $\mathbb{P}(A) > 0$.
- ▶ Bayes rule: $P(B/A) = \frac{P(A/B)P(B)}{P(A)}$.

Independence and Mutually exclusive

- ▶ Two events A, B are independent iff $P(A/B) = P(A)$ and $P(B/A) = P(B)$.
- ▶ Two events A, B are independent iff $P(A \cap B) = P(A)P(B)$.

- ▶ If A and B are independent, then so are A^c and B^c .
- ▶ What about A and B^c ? Are they independent?
- ▶ Two events A and B are mutually exclusive if occurrence of one implies that the other event cannot occur. Are they independent?
- ▶ If A and B are mutually exclusive, then they are not independent (and vice versa).

Random variable

- ▶ Given a random experiment with associated $(\Omega, \mathcal{F}, \mathbb{P})$, it is sometimes difficult to deal directly with $\omega \in \Omega$. eg. rolling a dice ten times.
- ▶ Notice that each sample point $\omega \in \Omega$ is not a number but a sequence of numbers.
- ▶ Also, we may be interested in functions of these sample points rather than samples themselves. eg: Number of times 6 appears in the 10 rolls.
- ▶ In either case, it is often convenient to work in a new *simpler* probability space rather than the original space.
- ▶ Random variable is a device which precisely helps us make this mapping from $(\Omega, \mathcal{F}, \mathbb{P})$ to a 'simpler' $(\Omega', \mathcal{F}', P_X)$.
- ▶ P_X is called as an induced probability measure on Ω' .

Random variable

- ▶ If Ω' is countable, then the random variable is called a discrete random variable.
- ▶ In this case it is convenient to use $\mathcal{F}'$ as power-set.
- ▶ If $\Omega' \subseteq \mathbb{R}$ or uncountable, then the random variable is a continuous random variable.
- ▶ In this case, $\mathcal{F}' = \mathcal{B}(\mathbb{R})$.
- ▶ Notation: Random variables denoted by capital letters like X, Y, Z etc. and their realizations by small letters x, y, z ..

PMF and CDF of a Discrete r.v.

- ▶ Let $X : \Omega \rightarrow \Omega'$ be a discrete r.v.
- ▶ Let $p_X(x)$ for $x \in \Omega'$ denote the probability that X takes the value x .
- ▶ $p_X(x)$ is called as a probability mass function.
- ▶ The cumulative distribution function (CDF) $F_X(\cdot)$ is defined as $F_X(x_1) := \sum_{x \leq x_1} p_X(x) = \mathbb{P}\{\omega \in \Omega : X(\omega) \leq x_1\}$.

Expectation, Moments, Variance

- ▶ The mean or expectation of a random variable X is denoted by $E[X]$ and is given by $E[X] = \sum_{x \in \Omega'} xp_X(x)$.
- ▶ The n^{th} moment of a random variable X is denoted by $E[X^n]$ and is given by $E[X^n] = \sum_{x \in \Omega'} x^n p_X(x)$.
- ▶ Functions of random variables are random variables.
- ▶ For a function $g(\cdot)$ of a random variable X , its expectation is given by $E[g(X)] := \sum_{x \in \Omega'} g(x)p_X(x)$
- ▶ $Var(X) := E[(X - E[X])^2]$
- ▶ HW: Prove that $E[(X - E[X])^2] = E[X^2] - E[X]^2$
- ▶ For $Y = aX + b$, what is $E[Y]$? $E[Y] = aE[X] + b$.
(Linearity of expectation)

Bernoulli random variable

- ▶ Bernoulli random variable $X = \begin{cases} 1, & \text{with probability } p \\ 0, & \text{otherwise.} \end{cases}$
- ▶ Basic models of Multi-arm bandit problem assume Bernoulli Bandits.
- ▶ $E[X] = p, E[X^n] = p.$

Binomial $B(n, p)$ random variable.

- ▶ Consider a biased coin (head with probability p) and toss it n times.
- ▶ Denote head by 1 and tail by 0.
- ▶ Let random variable N denote the number of heads in n tosses.
- ▶ PMF of N ?. $P_N(k) = \binom{n}{k} p^k (1 - p)^{n-k}$.
- ▶ HW: What is $E[N]$, $E[N^2]$, $\text{Var}(X)$?

Geometric random variable

- ▶ Consider a biased coin (head with probability p) and suppose you keep tossing it till head appears the first time.
- ▶ Let random variable N denote the number of tosses needed for head to appear first time.
- ▶ What is the PMF of N ? $p_N(k) = (1 - p)^{k-1}p$.
- ▶ HW: What is $E[N]$, $E[N^2]$, $Var(N)$?

Poisson random variable

- ▶ A Poisson random variable X comes with a parameter λ and has $\Omega' = \mathbb{Z}_{\geq 0}$
- ▶ PMF: $p_X(k) = e^{-\lambda} \frac{\lambda^k}{k!}$
- ▶ Intuitively its a limiting case of the Binomial distribution with n increasing and p decreasing such that np converges to λ .
- ▶ Mean of binomial is np so p should decrease while n increases.

Continuous random variables

- ▶ A random variable X is continuous if there exists a non-negative real valued probability density function (PDF) $f_X(\cdot)$ such that $F_X(x) = \int_{u=-\infty}^x f_X(u)du$.
- ▶ $P_X(a \leq X \leq b) = \int_a^b f_X(u)du$. (Area under the curve)

$$\frac{dF_X(x)}{dx} = f_X(x) \text{ or } P_X(x < X \leq x + h) \simeq f_X(x)h.$$

Mean, Variance, Moments

- ▶ $E[X] = \int_{-\infty}^{\infty} uf_X(u)du$
- ▶ $E[X^n] = \int_{-\infty}^{\infty} u^n f_X(u)du$
- ▶ $E[g(X)] = \int_{-\infty}^{\infty} g(u)f_X(u)du$
- ▶ $\text{Var}[X] = E[g(X)]$ where $g(x) = (x - E[X])^2$.
- ▶ For $Y = aX + b$, $E[Y] = aE[X] + b$.

Exponential random variable ($Exp(\lambda)$)

- ▶ This is a non-negative r.v. with parameter λ .
- ▶ Its pdf $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$.
- ▶ Its CDF is given by $F_X(x) = 1 - e^{-\lambda x}$ for $x \geq 0$.
- ▶ $E[X] = \frac{1}{\lambda}$ and $Var(X) = \frac{1}{\lambda^2}$
- ▶ $E[X^n] = \frac{n!}{\lambda^n}$

Summary: Multiple random variables

$$p_{XY}(x, y) := \mathbb{P}\{\omega \in \Omega : X(\omega) = x \text{ and } Y(\omega) = y\}.$$

$$F_{XY}(x, y) := \mathbb{P}\{\omega \in \Omega : X(\omega) \leq x \text{ and } Y(\omega) \leq y\}.$$

The marginal PMF's p_X and p_Y can be obtained from the joint PMF as follows:

$$p_X(x) = \sum_y p_{XY}(x, y) \text{ and } p_Y(y) = \sum_x p_{XY}(x, y).$$

Two random variables, X and Y are independent if the following is true:

$$p_{XY}(x, y) = p_X(x)p_Y(y), F_{XY}(x, y) = F_X(x)F_Y(y) \text{ and } E[XY] = E[X]E[Y].$$

$$E[g(X, Y)] = \sum_{xy} g(xy)p_{XY}(xy)$$

The rules for continuous random variables are similar.
Also revise conditioning of variables.

Sums of independent random variable

- ▶ Consider $Z = X + Y$. What is the pdf of Z when X and Y ?
- ▶ What is $p_Z(z)$ or $f_Z(z)$?
- ▶ $p_Z(z) = \sum_{(x,y):x+y=z} p_{X,Y}(x,y)$
- ▶ $f_Z(z) = \int_{(x,y):x+y=z} f_{X,Y}(x,y) dx dy$.
- ▶ Since X and Y are independent $p_{X,Y}(x,y) = p_X(x)p_Y(y)$ and $f_{X,Y}(x,y) = f_X(x)f_Y(y)$. This gives us

Convolution formula

$$p_Z(z) = \sum_x p_X(x)p_Y(z-x)$$
$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)f_Y(z-x)dx$$

HW: What if X and Y are not independent?

MGF of Sums of independent random variable

- ▶ Consider $Z = X + Y$. What is the pdf of Z when X and Y ?
- ▶ Let $M_X(t)$ and $M_Y(t)$ be their MGF's. What is $M_Z(t)$?
- ▶ $M_Z(t) = E[e^{Zt}] = E[e^{(X+Y)t}]$.
- ▶ $M_Z(t) = E[e^{Xt}.e^{Yt}]$.
- ▶ If X and Y are independent, $E[XY] = E[X]E[Y]$ and $E[g(X)h(Y)] = E[g(X)]E[h(Y)]$.
- ▶ $M_Z(t) = E[e^{Xt}].E[e^{Yt}]$.

$$M_Z(t) = M_X(t)M_Y(t).$$

MGF of Sums of independent random variable

- ▶ Consider $Z = X + Y$. What is the MGF of Z when X and Y ?

$$M_Z(t) = M_X(t)M_Y(t).$$

- ▶ What about $M_Z(t)$ when $Z = X_1 + X_2 + \dots X_n$ and X_i are iid.?
- ▶ $M_Z(t) = (M_X(t))^n$.
- ▶ What about $M_Z(t)$ when $Z = X_1 + X_2 + \dots X_N$ where N is a positive discrete random variable? section 4.5

Convergence of Random Variables

Summary

Pointwise
convergence

$$\lim_{n \rightarrow \infty} X_n(\omega) = X(\omega) \text{ for every } \omega$$

Almost sure
convergence

$$\lim_{n \rightarrow \infty} X_n(\omega) = X(\omega) \text{ almost surely}$$

Convergence
in probability

$$\lim_{n \rightarrow \infty} P(|X_n - X| > \epsilon) = 0 \text{ for any } \epsilon > 0$$

Mean-square
convergence

$$\lim_{n \rightarrow \infty} E[(X_n - X)^2] = 0$$

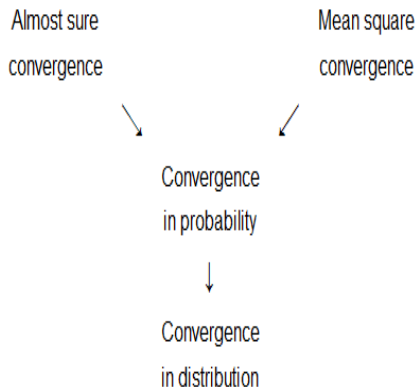
Convergence
in distribution

$$\lim_{n \rightarrow \infty} F_n(x) = F(x) \text{ for any continuity point } x$$

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¹Image from probabilitycourse.com

Relation between modes of convergence (no proofs)



https://en.wikipedia.org/wiki/Proofs_of_convergence_of_random_variables

Introduction to Stochastic processes

Introduction to Stochastic processes

- ▶ Stochastic process $\{X(t), t \in T\}$ on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a collection of random variables defined such that for every $t \in T$ we have $X(t) : \Omega \rightarrow \mathcal{S}$.
- ▶ T is the parameter space (often resembles time) and $\mathcal{S}$ is the state space.
- ▶ When T is countable, we have a discrete time process. If T is a subset of real line, we have a continuous time process.
- ▶ State space could be integers or real numbers
- ▶ State space could even be $\mathbb{R}^n$ or $\mathbb{Z}^n$ valued

Examples of Stochastic Processes

- ▶ Sequence of i.i.d random variables.
- ▶ General random walk: If $X_1, X_2, \dots$ is a sequence i.i.d of random variables, then $S_n = \sum_{i=1}^n X_i$ is a random walk.
- ▶ Wiener process: $\{X(t), t \geq 0\}$ is a Wiener process if
 1. $X(0) = 0$
 2. $\{X(t), t \geq 0\}$ has **stationary and independent increments**
 3. for every $t > 0$, $X(t)$ is normally distributed with mean 0 and variance t .
- ▶ $\{X(t), t \geq 0\}$ is a Markov process if for $t_1 < t_2 < \dots t_n < t$ we have
$$P(X(t) \leq x | X(t_1) = x_1, \dots, X(t_n) = x_n) = P(X(t) \leq x | X(t_n) = x_n)$$
- ▶ Random walk and Wiener process are examples of Markov processes.

Introduction to Stochastic processes

A c.t.s.p. is called an *independent increment process* if for any choice of parameters $t_0 < t_1 < \dots < t_n$, the n increment random variables $X(t_1) - X(t_0), X(t_2) - X(t_1), \dots, X(t_n) - X(t_{n-1})$ are independent.

The c.t.s.p. is said to have *stationary increments* if in addition $X(t_2 + s) - X(t_1 + s)$ has the same distribution as $X(t_2) - X(t_1)$ for all $t_1, t_2 \in T$ and any $s > 0$.

Counting process

Stochastic process $\{N(t), t \geq 0\}$ is a counting process if it represents the total number of events upto time t .

It satisfies the following

- ▶ $N(t) \geq 0$ and is integer valued
- ▶ For $s \leq t$, we have $N(s) \leq N(t)$. $N(t) - N(s)$ denotes the number of events in the interval (t, s)
- ▶ We will now look at some examples of counting processes that have independent and stationary increments.